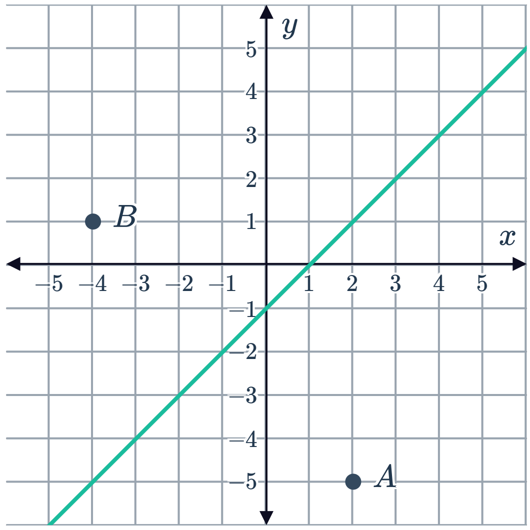
Add a site to a Voronoi diagram

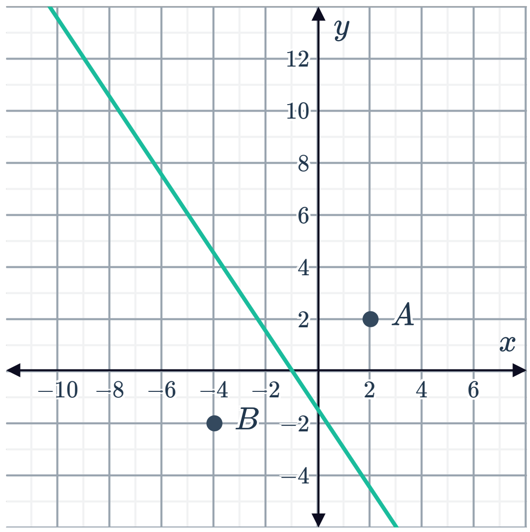
1 Consider the following Voronoi diagram for sites A and B:



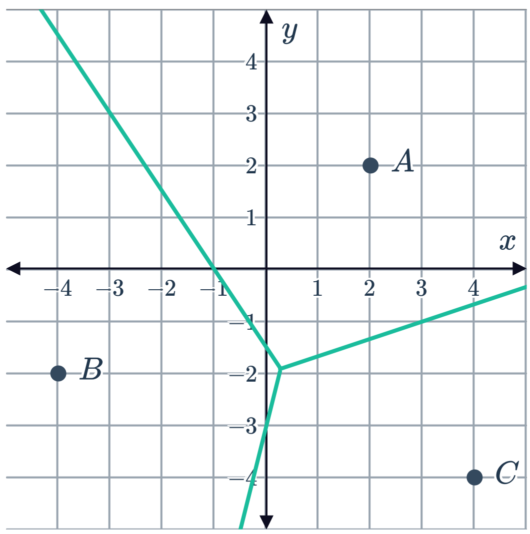
a Describe how you think adding the site $C(4, 3)$ to the Voronoi diagram would change it.

b Redraw the Voronoi diagram to include site $C(4, 3)$.

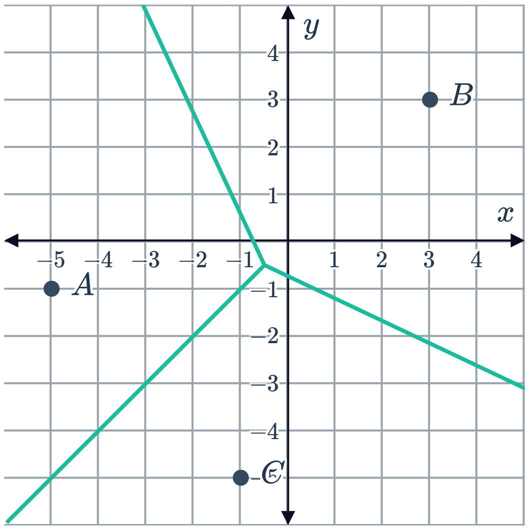
2 Add the site $C(0, 0)$ to the following Voronoi diagram:



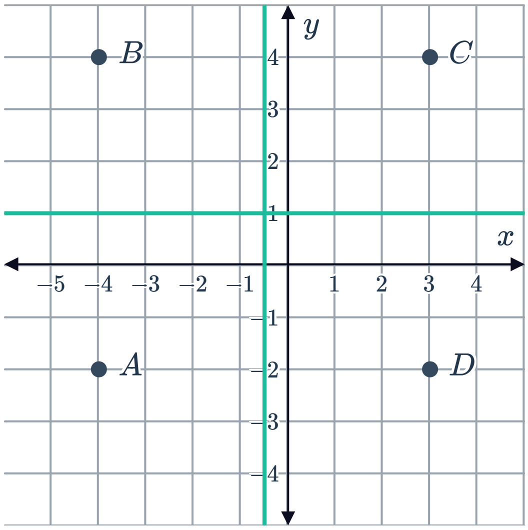
3 Add the site $D(-2, 2)$ to the following Voronoi diagram:



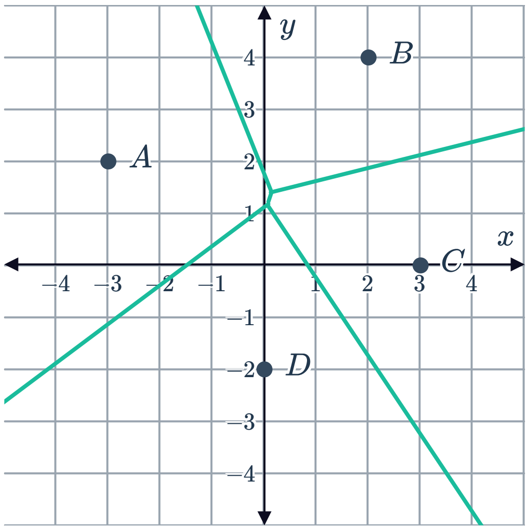
4 Add the site $D(-1, -1)$ to the following Voronoi diagram:



5 Add the site $E(-2, -4)$ to the following Voronoi diagram:

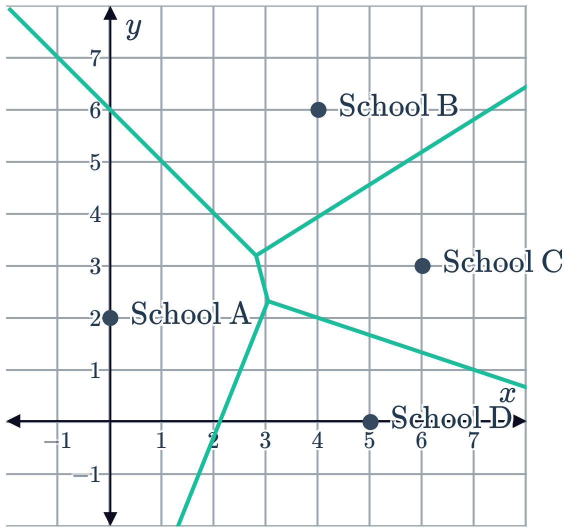


6 Add the site $E(-3, -4)$ to the following Voronoi diagram:



Applications

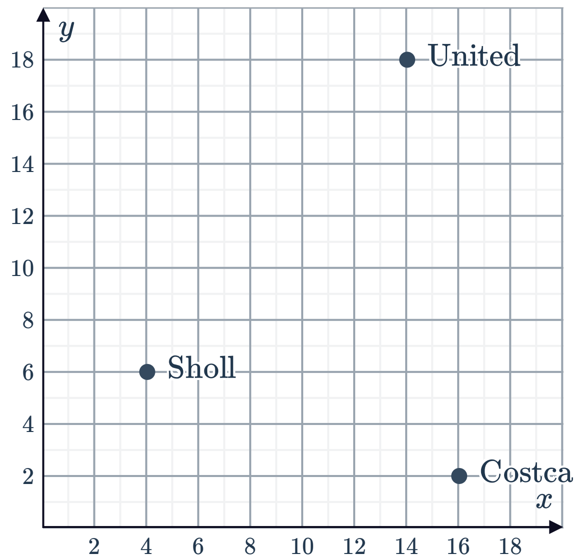
7 The following Voronoi diagram shows the locations of four schools in a city. Families are encouraged to send their children to their local (nearest) school. Each unit on the graph represents $1\ \mathrm{km}$.
A new school, School E has just been built at the location $(-1, 6)$.



a Add School E to the Voronoi diagram.

b Aoife lives at $(0, 5)$, and had been planning to attend School A next year. Should she still attend that school? Explain your answer.

8 The following map shows the locations of petrol stations in a city. Units are in kilometres.



a Construct a Voronoi diagram for these three petrol stations.

b A new petrol station called DP is about to be built. The owners of the other three stations are concerned that it will be too close to them and be bad for business. It is decided that the new petrol station must be at least $8\ \mathrm{km}$ away from the other petrol stations and be equidistant to exactly two of the current petrol stations.
Choose a location for the new petrol station, DP, that satisfies these requirements.

c Add the petrol station, DP, to the Voronoi diagram at your chosen location.